

A high-resolution photograph taken from space, showing the Earth's horizon as a bright blue arc against the black void of space. Below the horizon, the Earth's surface is covered in a dense layer of white clouds. A portion of a spacecraft, likely the International Space Station, is visible in the foreground, featuring a white cylindrical module and a long, metallic arm extending upwards. The lighting is bright, suggesting the sun is high in the sky.

First year Aerospace Engineering

Space II

Marcos Carrera Valle

Orbital Mechanics

- We want to know the position (and velocity) of a satellite at any instant of time
- Because the trajectory of a satellite is determined by its initial position and velocity after launch

Fundamentals

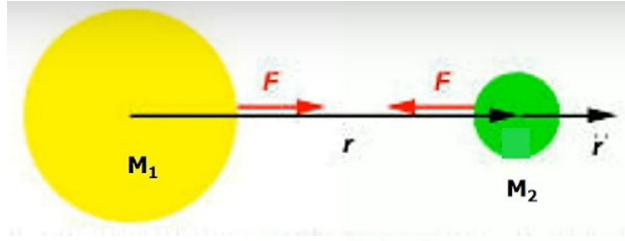
Kepler's Laws of Planetary Motion:

1. The orbits of the planets are ellipses, with the Sun at one focus of the ellipse
2. The line joining a planet to the sun sweeps out equal areas in equal times as the planet travels around the ellipse
3. The ratio of the squares of the revolutionary periods of two planets is equal to the ratio of the cubes of their semi-major axis

Newton's 3 laws of motion



Newton's Law of Universal Gravitation



Newton's Law of Universal Gravitation:

Every point mass attracts every other point mass by a force pointing along the line connecting both points. The force is directly proportional to the product of the two masses and inversely proportional to the square of the distance between the point masses:

$$F = G \frac{M_1 M_2}{r^2}$$

$$F = G \frac{M_1 M_2}{r^2} = M_1 a_1 = M_2 a_2$$

Newton's Law of Universal Gravitation

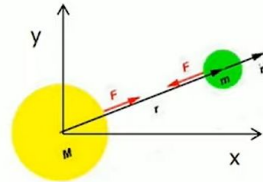
Relative acceleration between object 1 and object 2:

$$\ddot{\mathbf{r}} = -\mathbf{a}_2 - \mathbf{a}_1 = -\frac{\mathbf{F}}{M_2} - \frac{\mathbf{F}}{M_1} = -G \frac{M_1 + M_2}{r^2} \mathbf{\hat{r}}$$

$$\ddot{\mathbf{r}} = -G \frac{M_1}{r^2} \mathbf{\hat{r}} = -\frac{\mu}{r^2} \mathbf{\hat{r}}$$

Conversion to 3D case (vector notation):

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^2} \frac{\mathbf{r}}{r} = -\frac{\mu}{r^3} \mathbf{r} \quad \text{or} \quad \begin{pmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{pmatrix} = -\frac{\mu}{r^3} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$



Numerical example acceleration

Question:

Consider the Earth ($GM = \mu = 398600.441 \text{ km}^3/\text{s}^2$; $R_e = 6378.136 \text{ km}$).
What is (the absolute value of) the gravitational acceleration?

1. at sea surface
2. for an earth-observation satellite at 800 km altitude
3. for a GPS satellite at 20,200 km altitude
4. for a geostationary satellite at 35,800 km altitude

Numerical example acceleration (2)

1. Consider the situation of the Earth, the Sun and a satellite somewhere on the line connecting the two main bodies. Where is the point where the direct attracting forces of Earth and the Sun, acting on the satellite, are in equilibrium?

Data: $\mu_{\text{Earth}} = 398600.441 \text{ km}^3/\text{s}^2$, $\mu_{\text{Sun}} = 1.327178 \times 10^{11} \text{ km}^3/\text{s}^2$,
1 AU (average distance Earth-Sun) = $149.6 \times 10^6 \text{ km}$.

Hint: make general formulation, and solve by trial-and-error.

2. The Moon orbits Earth at a distance w.r.t. the center-of-mass of Earth of about 384,000 km. Still, the Sun does not pull it away from Earth. Why not?

Fundamentals: conservation of angular momentum

1) Vectorial product of equation of motion with $\mathbf{r}$:

$$\mathbf{r} \times \ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r} \times \mathbf{r} = 0$$

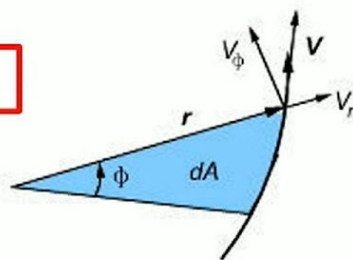
so

$$\mathbf{r} \times \ddot{\mathbf{r}} = \frac{d}{dt}(\mathbf{r} \times \dot{\mathbf{r}}) - \dot{\mathbf{r}} \times \dot{\mathbf{r}} = \frac{d}{dt}(\mathbf{r} \times \dot{\mathbf{r}}) = 0$$

so

$$\mathbf{r} \times \dot{\mathbf{r}} = \boxed{\mathbf{r} \times \mathbf{V} = \text{constant} = \mathbf{H}}$$

- the motion is in one plane
- $H = r V_\phi = r (r d\phi/dt) = r^2 d\phi/dt = \text{constant}$
- area law (**second Law of Kepler**): $dA/dt = \frac{1}{2} r V_\phi = \frac{1}{2} r r d\phi/dt = \frac{1}{2} H$



Fundamentals

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Newton's 3 laws of motion



Fundamentals: conservation of energy

2) Scalar product of equation of motion with $d\mathbf{r}/dt$:

$$\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} + \frac{\mu}{r^3} \dot{\mathbf{r}} \cdot \mathbf{r} = 0$$

$$\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} = \frac{1}{2} \dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} + \frac{1}{2} \ddot{\mathbf{r}} \cdot \dot{\mathbf{r}} = \frac{d}{dt} \left(\frac{1}{2} \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} \right) \quad \dot{\mathbf{r}} \cdot \mathbf{r} = \frac{1}{2} \dot{\mathbf{r}} \cdot \mathbf{r} + \frac{1}{2} \mathbf{r} \cdot \dot{\mathbf{r}} = \frac{d}{dt} \left(\frac{1}{2} \mathbf{r} \cdot \mathbf{r} \right)$$

$$\frac{1}{2} \frac{d}{dt} (\dot{\mathbf{r}} \cdot \dot{\mathbf{r}}) + \frac{1}{2} \frac{\mu}{r^3} \frac{d}{dt} (\mathbf{r} \cdot \mathbf{r}) = \frac{1}{2} \frac{d}{dt} (V^2) + \frac{1}{2} \frac{\mu}{r^3} \frac{d}{dt} (r^2) = 0$$

$$\frac{1}{2} \frac{d}{dt} (V^2) + \frac{1}{2} \frac{\mu}{r^3} 2r \dot{r} = 0$$

$$\frac{d}{dt} \left(\frac{V^2}{2} \right) - \frac{d}{dt} \left(\frac{\mu}{r} \right) = 0$$

Now, $\frac{d}{dt} \left(\frac{\mu}{r} \right) = -\frac{\mu}{r^2} \cdot \dot{r}$, so

$$\frac{\mu}{r^2} \dot{r} = -\frac{d}{dt} \left(\frac{\mu}{r} \right)$$

Fundamentals: conservation of energy

$$\frac{d}{dt}\left(\frac{V^2}{2}\right) - \frac{d}{dt}\left(\frac{\mu}{r}\right)$$

$$\frac{d}{dt}\left(\frac{1}{2}V^2 - \frac{\mu}{r}\right) = 0$$

Integration: $\frac{V^2}{2} - \frac{\mu}{r} = \text{constant} = E$

$\nearrow$ kinetic energy $\nwarrow$ potential energy
(incl. minus sign!)

$$\frac{V^2}{2} = \frac{\frac{1}{2}mV^2}{m} = \frac{K.E.}{m}$$

$$-\frac{\mu}{r} = g r \quad \left(g = -\frac{\mu}{r^2}\right)$$
$$g r = \frac{m g r}{m} = \frac{P.E.}{m}$$

Fundamentals: orbit equation

3) Scalar product of equation of motion with $\mathbf{r}$:

$$\mathbf{r} \cdot \ddot{\mathbf{r}} + \frac{\mu}{r^3} \mathbf{r} \cdot \mathbf{r} = 0$$

Therefore:

$$\frac{d}{dt}(\mathbf{r} \cdot \dot{\mathbf{r}}) - (\dot{\mathbf{r}} \cdot \dot{\mathbf{r}}) + \frac{\mu}{r} = 0$$

$$\begin{aligned} \frac{d}{dt}(\mathbf{r} \cdot \dot{\mathbf{r}}) &= \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} + \mathbf{r} \cdot \ddot{\mathbf{r}} \\ \hookrightarrow \mathbf{r} \cdot \ddot{\mathbf{r}} &= \frac{d}{dt}(\mathbf{r} \cdot \dot{\mathbf{r}}) - \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} \end{aligned}$$

Note that $\mathbf{r} \cdot \dot{\mathbf{r}} = \mathbf{r} \cdot \mathbf{V} = r V_r = r \dot{r}$ and $\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} = \mathbf{V} \cdot \mathbf{V} = V^2$

$$\frac{d}{dt}(r\dot{r}) = r\ddot{r} + \dot{r}\dot{r} = r\ddot{r} + \dot{r}^2$$

so:

$$r\ddot{r} + \dot{r}^2 - V^2 + \frac{\mu}{r} = 0$$

Substitution of
yields:

$$V^2 = V_r^2 + V_\phi^2 = \dot{r}^2 + (r\dot{\phi})^2$$

$$\ddot{r} - r\dot{\phi}^2 = -\frac{\mu}{r^2}$$

Fundamentals: orbit equation (cnt'd)

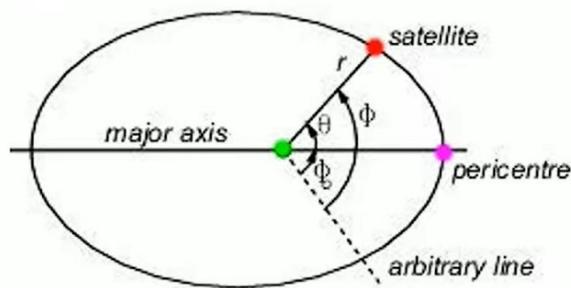
Combining equations $r^2 \dot{\phi} = H$ and $\ddot{r} - r\dot{\phi}^2 = -\frac{\mu}{r^2}$

gives the equation for a conical section (**first Law of Kepler**):

$$r = \frac{p}{1 + e \cos(\theta)}$$

where

- $\theta = \phi - \phi_0$ = true anomaly
- ϕ_0 = arbitrary angle
- e = eccentricity
- $p = H^2/\mu$ = semi-latus rectum



Elliptical orbit

Orbital equation:

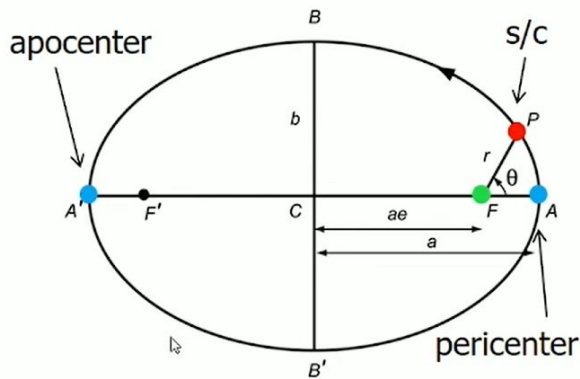
$$2a = r_{\theta=0} + r_{\theta=\pi} = \frac{p}{1+e} + \frac{p}{1-e} = \frac{p}{1+e} \frac{1-e}{1-e} + \frac{p}{1-e} \frac{1+e}{1+e} = \frac{2p}{1-e^2} \Rightarrow p = a(1-e^2)$$

so:

$$r = \frac{a(1-e^2)}{1+e\cos(\theta)}$$

Other expressions:

- pericenter A distance $r_p = a(1-e)$
- apocenter A' distance $r_a = a(1+e)$
- semi-major axis $a = (r_a + r_p) / 2$
- eccentricity $e = (r_a - r_p) / (r_a + r_p)$
- location of focal center $CF = a - r_p = ae$



after [Wakker, 2015]

Questions

According to Kepler's First Law, what is the shape of a planetary orbit?

- a) Circular
- b) Elliptical
- c) Parabolic
- d) Hyperbolic

Newton's Law of Universal Gravitation states that the gravitational force between two objects is:

- a) Directly proportional to their masses and inversely proportional to the square of the distance between them
- b) Directly proportional to the square of their masses and inversely proportional to their distance
- c) Independent of the distance between the objects
- d) Only dependent on the larger object's mass

Questions

Which of the following best describes Kepler's Second Law?

- a) The ratio of the squares of the orbital periods of planets is equal to the ratio of the cubes of their semi-major axes
- b) The orbits of planets are elliptical, with the Sun at one focus
- c) A line connecting a planet to the Sun sweeps out equal areas in equal time intervals
- d) Planets further from the Sun move faster than those closer

What does Kepler's Third Law allow us to determine?

- a) The speed of a planet at any given moment
- b) The relationship between the period of orbit and the semi-major axis of the orbit
- c) The gravitational force acting on the planet
- d) The eccentricity of the orbit

Questions

Two satellites orbiting Earth have masses of 500 kg and 1,000 kg respectively. If the distance between them is 200 km, what happens to the gravitational force if the distance doubles?

- a) It doubles
- b) It quadruples
- c) It halves
- d) It becomes one-fourth

Questions

The Moon orbits Earth at an average distance of 384,400 km with an orbital period of 27.3 days. If another moon were discovered orbiting Earth with twice this distance, what would be its approximate orbital period?

- a) 38.6 days
- b) 54.6 days
- c) 77.1 days
- d) 96.9 days

Questions

A satellite orbits Earth with a semi-major axis of 10,000 km and an eccentricity of 0.2. What is its closest approach to Earth (perigee distance)?

- a) 8,000 km
- b) 10,000 km
- c) 12,000 km
- d) 14,000 km

Questions

Two planets, A and B, orbit a star. Planet A has an orbital period of 2 years, while Planet B's semi-major axis is twice that of Planet A. What is the orbital period of Planet B?

- a) 4 years
- b) 5.7 years
- c) 8 years
- d) 11.3 years

Questions

A planet orbits a star in an elliptical orbit with an eccentricity of 0.5 and a semi-major axis of 150 million km. What is its farthest distance from the star (apogee)?

- a) 100 million km
- b) 150 million km
- c) 200 million km
- d) 225 million km

Questions

If the mass of the Sun were doubled while keeping Earth's distance the same, how would Earth's orbital period change?

- a) It would increase
- b) It would decrease
- c) It would remain the same
- d) It would become unpredictable

Questions

A planet is observed to have an elliptical orbit with an eccentricity of 0.8 and a semi-major axis of 500 million km. What is the planet's closest distance to its star?

- a) 100 million km
- b) 200 million km
- c) 300 million km
- d) 400 million km

Questions

Two satellites, A and B, orbit the Earth. Satellite A orbits at 10,000 km from the Earth's center, while Satellite B orbits at 40,000 km. How does Satellite B's period compare to Satellite A's?

- a) 2 times longer
- b) 4 times longer
- c) 8 times longer
- d) 16 times longer

Questions

A spacecraft in orbit around Earth suddenly increases its velocity at perigee. How does this affect its orbit?

- a) The orbit becomes more circular
- b) The orbit becomes more elliptical with a higher apogee
- c) The spacecraft moves to a lower orbit
- d) The eccentricity of the orbit decreases

Questions

A satellite is in a circular orbit around Earth with a radius of 10,000 km. If it moves to an elliptical orbit with an apogee of 20,000 km, how does its velocity at perigee compare to its original velocity?

- a) Increases
- b) Decreases
- c) Remains the same
- d) Becomes unpredictable

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